

Predicting muscle anaerobic threshold by using a wearable optical sensor during exercise

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ABSTRACT

Background: The anaerobic threshold (AT) is a point during intense exercise that can be used to predict muscular fatigue. Determining the AT non-invasively helps to adjust exercise intensity and prevent overuse injuries. Near-infrared spectroscopy (NIRS) is an optical technology that can provide real-time information about muscle oxidative metabolism. The objective of this pilot study was to investigate the relationship between NIRS parameters of muscle oxygenation and traditional measures of exercise monitoring, such as heart rate and relative body oxygen consumption (VO_2).

Methods: Healthy adults with moderate to high fitness levels participated in an incremental exercise protocol on a stationary bicycle. NIRS parameters were compared to ventilatory VO_2 using a metabolic cart. Respiratory Exchange Ratio (RER) > 1.0 was used as a proxy for determining the AT. NIRS data were collected from the primary locomotor muscle (vastus lateralis - VL) and a control muscle (deltoid) using two wearable NIRS sensors. Heart rate data were collected by a wearable ECG sensor.

Results: The NIRS data showed a significant decline in VL muscle oxygenated hemoglobin (O_2Hb) concentration ($p < 0.05$) at one exercise stage after the AT was identified. Muscle O_2Hb did not show a significant decrease in the deltoid at the AT. Furthermore, there were no noticeable changes in heart rate at the AT.

Conclusion: Our results indicate that a wearable NIRS sensor can predict the AT in exercising muscles and may provide a localized measure of muscular fatigue during exercise.

Keywords: wearable sensor; muscle oxygenation; near-infrared spectroscopy; VO_2 ; exercise monitoring; anaerobic threshold; sports sensor; muscular fatigue

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1. INTRODUCTION

The anaerobic threshold (AT), also known as the lactate threshold, is a point during exercise where the production of lactate in the blood is equal to the rate of blood lactate removal.¹ Continuing to exercise past the AT can significantly impact cardiopulmonary performance during exercise.² The associated accumulation of blood lactate can contribute to impaired muscle contraction, hyperventilation, and oxygen delivery, which can negatively impact athletic performance.² As a result, the AT is a key predictor of aerobic endurance and can be a useful tool for athletes and coaches to develop fitness levels and training plans. Prior to the AT, energy is produced through aerobic metabolism using oxygen. After the AT, energy is produced through anaerobic metabolism and results in a sharp increase in blood lactate concentration. The ability to produce energy using aerobic metabolism during exercise for longer is associated with higher aerobic efficiency and fitness levels.² Furthermore, by training at or slightly past the AT, it is possible to delay the accumulation of lactate and improve aerobic efficiency.² Therefore, the ability to detect the AT in real time during exercise can provide a significant advantage to athletes and coaches for developing aerobic fitness and preventing overtraining.

The gold standard for determining the AT is monitoring blood lactate concentration. During exercise, blood lactate is sampled repeatedly, and the AT is determined after an abrupt, non-linear rise in blood lactate concentration of at least 1mmol/L.³ The AT can also be estimated non-invasively by measuring respiratory gases continuously. There are several methods for determining the AT using respiratory gases, including the ratio of expired carbon dioxide (CO₂) to inspired oxygen (O₂), known as the respiratory exchange ratio (RER).³ Another simple and inexpensive way to estimate the AT is to monitor heart rate during exercise. During an incremental exercise protocol, exercise velocity and heart rate share a linear relationship. At the AT, the heart rate begins to increase less than the exercise velocity, and a deflection point is visible.⁴⁻⁵ While these three methods have shown success in determining the AT during exercise, each comes with several limitations. Primarily, these methods require specific testing conditions that make them difficult to implement in the field of play. As a result, there is a lack of agreement about which procedure is most effective for determining the AT. The ideal device for monitoring athletic performance is wearable, non-invasive, continuous, and reliable.

Near-infrared spectroscopy (NIRS) is an optical technology that uses light in the near-infrared (NIR) spectrum (650-1000 nm) to provide a continuous measurement of tissue oxygenation and hemodynamics. NIRS measures the changes in oxygenated (O₂Hb) and de-oxygenated (HHb) hemoglobin chromophore concentration, which can provide information about tissue oxygenation status in real time.⁶ The sum of O₂Hb and HHb is total hemoglobin (THb) and is a measure of tissue blood volume changes. NIRS sensors that contain multiple wavelengths of NIR light, or multiple source-detector distances in a spatially resolved (SR) configuration, can be used to provide an estimate of tissue oxygen saturation (TOI%).⁶⁻⁸ NIRS has demonstrated potential as a useful monitoring tool in musculoskeletal and sports medicine applications.^{6,9-11} By monitoring changes in muscle oxygenation, NIRS may be able to provide a reliable estimate of the AT and predict muscular fatigue during exercise. Furthermore, the non-invasive and continuous nature of NIRS has the potential to meet the criteria of an ideal sports monitoring device for high-performance athletes.

1.1 Objectives

The objective of this pilot study was to investigate the relationship between NIRS parameters of muscle oxygenation and traditional measures of monitoring athletic performance, such as heart rate and relative oxygen consumption during exercise. The outcomes of this study will contribute to developing a novel, reliable, and non-invasive biosensor to measure muscular fatigue based on the prediction of AT.

2. METHODS

In this study, we examined the ability of transcutaneous NIRS monitoring to collect muscle oxygenation signals and define the AT during an incremental exercise protocol on a stationary bicycle. The NIRS data were compared to other relevant physiological parameters, including RER and heart rate.

2.1 Participants

Eight young, healthy male adults were recruited for participation in the study. Participants were included in the study if they were healthy adults between the ages of 18-40 with moderate to high fitness levels. Fitness level was determined by a minimum of 3 hours of vigorous cardiovascular exercise per week. Participants were excluded from the study if they had a musculoskeletal injury affecting their ability to exercise, had a history of cardiopulmonary conditions or if they were taking any medication that affected their cardiopulmonary or metabolic response to exercise. The Clinical

Research Ethics Board at the University of British Columbia approved the study's protocol, and all participants provided informed consent at the beginning of a data collection session.

2.2 Instrumentation

This study used two compact, continuous-wave, wearable NIRS sensors (PortaLite/Portamon, Artinis Medical Systems, BV, Netherlands). Both sensors contained three light sources located at different distances from the photodetector, allowing for the acquisition of spatially resolved NIRS data. The light sources emitted NIR light at wavelengths of 760 nm and 850 nm. Data were collected at a sampling rate of 50 Hz and transmitted via Bluetooth[®] to a computer containing software for data visualization, analysis, and storage. The NIRS software allowed visualization of O₂Hb, HHb, THb, and TOI% in real time. A metabolic cart (TrueOne 2400, ParvoMedics, Sandy, UT) was used to provide a continuous analysis of respiratory gases throughout the exercise protocol. The metabolic cart provided a measure of oxygen consumption (VO₂), carbon dioxide production (VCO₂) and respiratory exchange ratio (RER). Respiratory data were recorded at an interval of 5s. Heart rate was monitored continuously using a wireless heart rate monitor (Frontier X2, Fourth Frontier, Austin, USA) placed over the chest. Heart rate data were collected at a sampling rate of 125 Hz. An electronically controlled stationary bicycle (SC3, Stages Cycling, Portland, USA) was used to complete the incremental exercise protocol. The stationary bicycle was connected to a training software (PerfPRO Studio 2021, Vision Quest Virtual, LLC, Chicago, USA), allowing the control of power output and the visualization of cadence and power.

2.3 Experimental Protocol

The study required a single visit to participate in an incremental exercise protocol from rest to task failure on an electronically controlled stationary bicycle. Prior to the exercise protocol, a trained research assistant placed each sensor in the appropriate location. The vastus lateralis muscle on the quadriceps was selected as the primary locomotor muscle during the stationary bicycle protocol. A NIRS sensor was affixed to the vastus lateralis using double-sided tape and held in place using a compression sleeve. The upper arm deltoid muscle was selected as the control muscle and monitored by a second NIRS sensor during the stationary bicycle protocol. The heart rate monitor was placed on the sternal manubrium and held in place using a flexible strap. The metabolic cart mask was then placed over the mouth of the participant, and the exercise test was ready to begin (Figure 1).



Figure 1. Experimental setup for the incremental exercise protocol.

The exercise protocol is illustrated in Figure 2. Participants began with a three-minute baseline sitting at rest on the stationary bicycle without pedalling. Following the baseline stage, participants began pedalling at a cadence of 70 rotations per minute (RPM) and a power output of 30W for five minutes. The five-minute exercise stage was followed by one minute of rest, where participants remained sitting on the bicycle without pedalling. After the rest period, participants continued cycling with a cadence of 70 RPM but an increased power output of 60W for five minutes. This was followed once again by one minute of rest. The protocol continued with five minutes of exercise followed by one minute of rest, with each exercise stage increasing in power output by 30W until task failure was reached. Task failure was reached when the participant stated they could not continue pedalling or if their cadence fell below 60 RPM. Following the exercise protocol, participants remained seated on the bike for 10 minutes to monitor the recovery period.

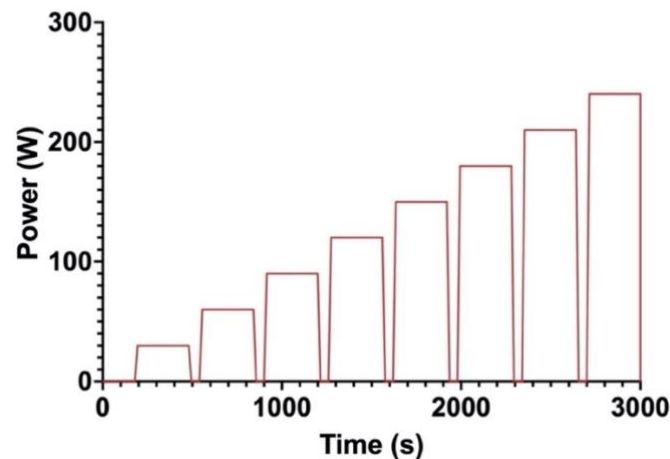


Figure 2. Power (W) vs. Time (s) graph describing the incremental exercise protocol.

2.4 Data Analysis

The NIRS, metabolic cart, and heart rate data were categorized into the last two minutes of each exercise step. The O₂Hb signal from the vastus lateralis was used as the primary output variable in this study. RER > 1.0 was used to estimate the AT.^{1,3} The normality of the data was tested using a Shapiro-Wilk test. A one-tailed Wilcoxon Signed Rank test was used to assess the differences in NIRS data between each exercise stage. Data analysis was carried out using MATLAB 2021.

3. RESULTS

NIRS, heart rate and ventilatory data were collected from eight participants during an incremental exercise protocol on a stationary bicycle. Mean participant characteristics are presented in Table 1.

Table 1: Characteristics of study participants

Age (y)	Height (cm)	Weight (kg)	VO _{2max} (mL·kg ⁻¹ ·min ⁻¹)	Power _{max} (W)	HR _{max} (bpm)
27.8 ± 4.1	177.6 ± 4.4	77.1 ± 5.7	37.8 ± 8.9	205.0 ± 35.1	173.6 ± 24.8

Mean ± standard deviation for the following variables: age, height, weight, maximum oxygen uptake (VO_{2max}), maximum power output (Power_{max}) and maximum heart rate (HR_{max}), N=8 for all characteristics

Seven participants reached the AT, as determined by the metabolic cart. In six of these participants, the NIRS data showed the same trend. During the initial stages of the exercise protocol, the O₂Hb signal from the vastus lateralis increased as power output increased. At the AT determined by RER > 1.0, there was a peak in the O₂Hb signal, and in the first exercise stage after the AT, there was a significant decrease ($p < 0.05$) in O₂Hb concentrations. The O₂Hb signal continued to decrease with each successive exercise stage after the AT. The results of the Wilcoxon Signed Rank test for the vastus lateralis muscle are presented in Table 2. Figure 3 displays an example of the RER and O₂Hb graph in the vastus lateralis during the exercise protocol.

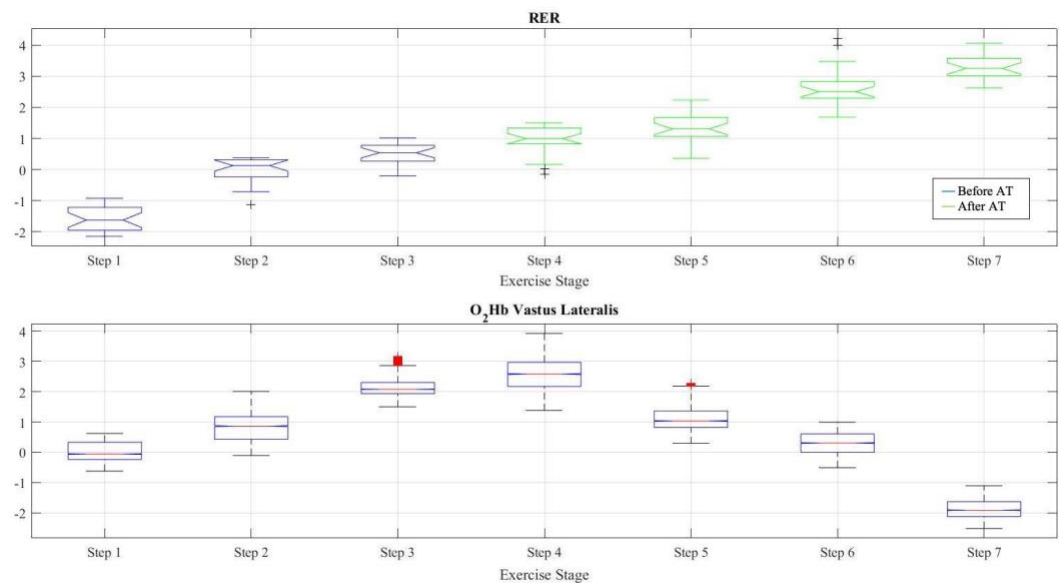


Figure 3. RER and O₂Hb graph from the vastus lateralis muscle from one participant.

In the deltoid muscle, there was no significant decrease in the O₂Hb signal after the AT ($p > 0.05$). This was consistent in all participants who reached the AT (n=7). The results of the Wilcoxon Signed Rank test for the deltoid muscle are presented in Table 2. Figure 4 displays an example of the RER and O₂Hb graph from the deltoid during the exercise protocol.

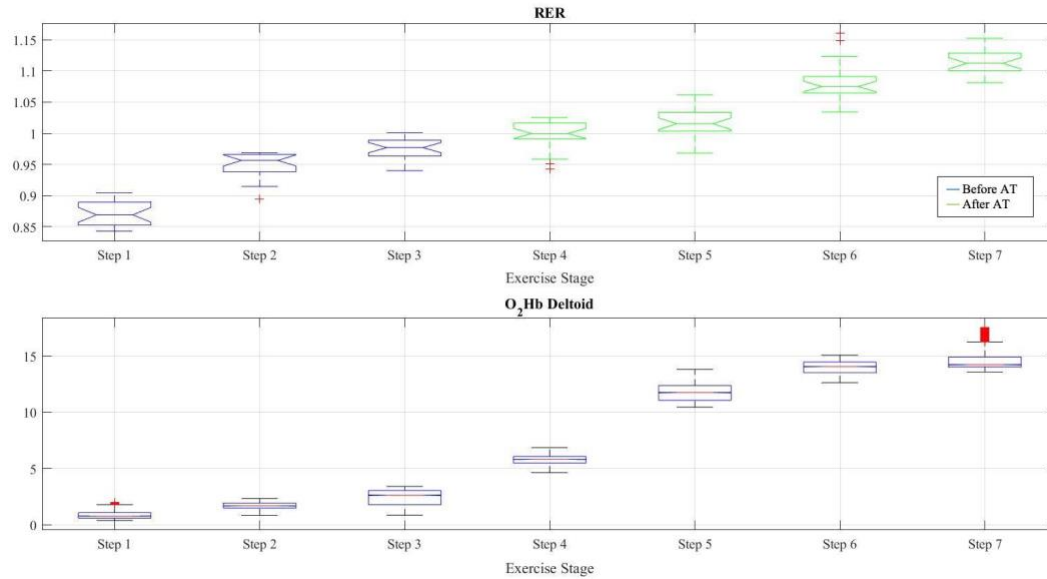


Figure 4. A representative figure of RER and deltoid muscle O₂Hb graphs in one participant.

Table 2: Results of the Wilcoxon Signed Rank test at the anaerobic threshold

	P01	P02	P03	P04**	P05	P06	P07	P08
Exercise Stage^a (AT)	5	8	4	-	4	3	4	3
p-value^b (VL)	0.00*	0.00*	0.00*	-	0.00*	0.00*	0.00*	0.00*
p-value^c (deltoid)	1.00	0.00*	1.00	-	1.00	1.00	1.00	0.43

^aExercise stage in which median RER > 1.0

^bp-value of Wilcoxon Signed Rank test between NIRS VL data at the AT and one exercise stage after the AT.

^cp-value of Wilcoxon Signed Rank test between NIRS deltoid data at the AT and one exercise stage after the AT.

*significantly less than, Note: $p < 0.05$

**P04 did not reach the anaerobic threshold

At the AT, there were no significant changes in heart rate for all participants ($p > 0.05$). An example of the RER and heart rate graphs for one participant is displayed in Figure 5.

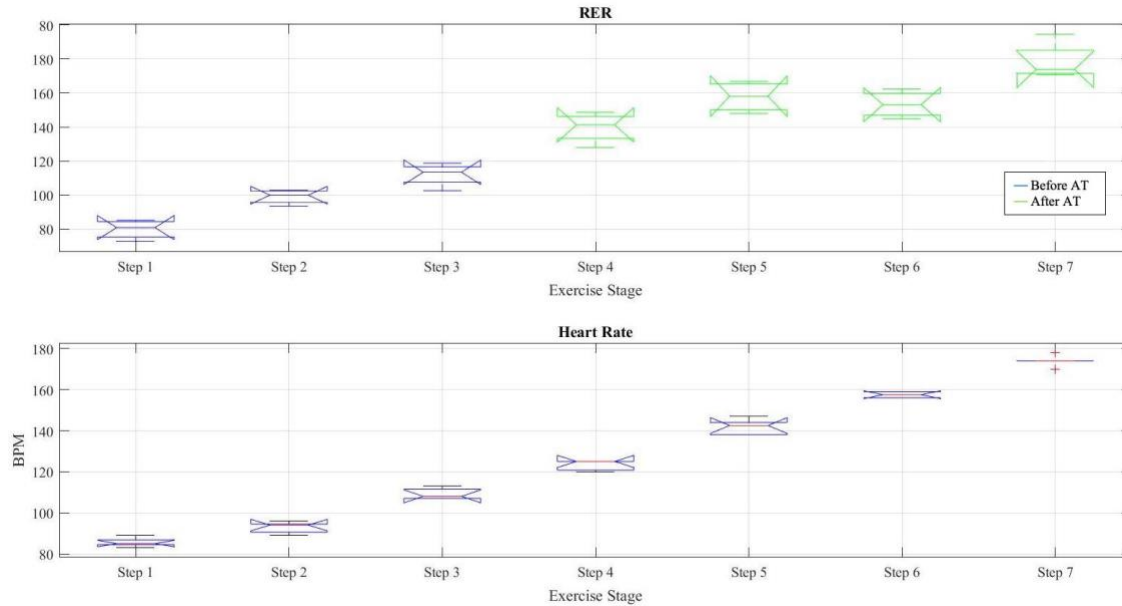


Figure 5. RER and heart rate graph from one participant.

4. DISCUSSION

In elite athletics, special care must be taken to build a training model that maximizes performance by applying an appropriate training load while also minimizing injury risk and fatigue.¹² A major contributing factor to athletic performance and athletic injury is the role of muscular fatigue, which is defined as the failure to maintain an expected force, intensity or power output.^{12,13} The accumulation of muscular fatigue can result in overtraining, which negatively impacts athletic performance.¹⁴ Furthermore, muscular fatigue has been associated with an increased risk of injury and illness in athletes.¹⁵ As a result, the ability to detect muscular fatigue in real time in the field of play would provide a significant advantage in training and competitive scenarios. The AT has been identified as a physiological metric that can be used as a predictor of muscular fatigue during exercise.² However, there has been difficulty in establishing a reliable method to detect the AT during exercise. Monitoring blood lactate is invasive and is not continuous, making it unreliable in the field of play. Additionally, readings can vary between arterial, capillary and venous blood.³ Ventilatory methods are expensive and require specific laboratory testing equipment that cannot be implemented in the field. Heart rate can be affected by physiological and environmental factors and is less effective in highly trained endurance athletes with lower resting heart rates.⁵ As a result, we currently lack a reliable method to determine the AT and monitor muscle metabolic fitness continuously and non-invasively in the field. Consequently, overuse and overtraining injuries are common in high-performance athletics.

By continuously monitoring muscle oxygenation and hemodynamics, near-infrared spectroscopy has the potential to meet the requirements of an ideal sports monitoring device that can be implemented in the field of play. In our study, a wearable NIRS sensor was affixed to the primary locomotor muscle (vastus lateralis) during an exercise protocol on a stationary bicycle. As participants began the exercise protocol, the O₂Hb signal in the NIRS data increased with each successive exercise stage. At the AT determined by ventilatory methods, there was a deflection point in the NIRS O₂Hb signal. At the exercise stage, when RER > 1.0, there was a peak in the O₂Hb data, followed by a decrease in O₂Hb in each successive exercise stage. This deflection point was consistent in six participants who reached the AT. In one of the participants, the O₂Hb signal from the vastus lateralis showed a deflection point two exercise stages before the AT. For the one participant that did not reach the AT, there was no deflection point in the NIRS O₂Hb signal. In the control muscle (deltoid), there was no deflection point in the NIRS O₂Hb signal, and the O₂Hb signal in the deltoid continued to increase for the duration of the exercise protocol. This may indicate that transcutaneous NIRS monitoring can predict the AT in

exercising muscles and provide a localized measure of muscular fatigue in real-time. When compared to ventilatory methods, there were no significant changes in the heart rate data at the AT for all participants.

The results of our study are consistent with research that has used NIRS to monitor exercising muscles. A similar study used a wearable NIRS sensor to show a decrease in O_2Hb in the primary locomotor muscle following the second ventilatory threshold, a physiological metric equal to the AT.¹⁶ Furthermore, several studies have described a plateau that occurs in the HHb signal of the primary locomotor muscle during incremental exercise.^{17,18} One study discovered a plateau that occurs in the HHb signal in both the primary locomotor muscle and non-locomotor muscle during incremental exercise.¹⁹ Technological improvements in the sports science field have aided in the process of implementing wearable devices and systems to monitor position, motion, biomechanics, impact and physiological status in the competition.¹² The abundance of recent NIRS research in the sports science field highlights the potential of NIRS as a useful tool in exercise monitoring. Therefore, developing and evaluating a wearable sensor containing a NIRS sensor to monitor exercising muscles continuously and non-invasively presents an exciting commercial opportunity.

We recognize limitations in our study, including our sample size being limited to 8 male participants. However, we are currently continuing data collection on a cohort with an increased sample size, including female participants and participants with varied levels of fitness, such as elite athletes. Furthermore, the use of a metabolic cart as our gold standard could only offer an estimate of the AT. To obtain a precise determination of the AT, lactate must be sampled repeatedly from the blood. However, we chose to use a continuous and non-invasive measure, which has been proven to provide a reliable estimate of the AT.^{1,3} Additionally, the results of our study were obtained during a specific exercise protocol. More research needs to be done to implement our findings in the field of play.

5. CONCLUSIONS

We demonstrated the ability of a wearable NIRS sensor to monitor muscle metabolic fitness continuously and non-invasively during incremental exercise and provide an estimate of the anaerobic threshold. Furthermore, we demonstrated that NIRS may provide a localized measure of muscular fatigue during exercise. Developing a wearable biosensor containing a NIRS sensor has the potential to become a useful modality to monitor exercising muscles and prevent overtraining injuries in high-performance athletics.

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